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Research article

RP-HPLC METHOD DEVELOPMENT AND VALIDATION FOR QUANTIFICATION OF ENTRECTINIB IN BULK AND PHARMACEUTICAL DOSAGE FORM

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ABSTRACT

A simple, rapid, specific and accurate reverse phase high performance liquid chromatographic method has been developed for the validated of Entrectinib in bulk as well as in marketed pharmaceutical dosage form. This separation was performed on a Symmetry ODS C18 (4.6×250mm, 5µm) column with Methanol: Phosphate Buffer (35:65) V/V as mobile phase at a flow rate of 1.0 mL min−1 with UV detection at 235 nm; the constant column temperature was Ambient. The runtime under these chromatographic conditions was less than 8 min. The retention time of Entrectinib was found to be 2.252. The calibration plot was linear over the concentration range of 6–14µg mL−1 with limits of detection and quantification values of 1.2 and 3.6ng mL−1 respectively. The mean % assay of marketed formulation was found to be 99.86%, and % recovery was observed in the range of 98-102%. Relative standard deviation for the precision study was found <2%. The developed method is simple, precise, specific, accurate and rapid, making it suitable for estimation of Entrectinib in bulk and marketed pharmaceutical dosage form dosage form.

Keywords: Entrectinib, RP-HPLC, Validation, ICH Guidelines

INTRODUCTION

Entrectinib N-[5-(3,5-Difluorobenzyl)-1H-indazol-3-yl]-4-(4-methyl-1- piperazinyl)-2- is an anti-cancer drug used to treat ROS1- positive non-small cell lung cancer and NTRK fusion-positive solid tumors. Entrectinib is a tyrosine kinase inhibitor; hence, it acts on several receptors. It acts as an adenosine triphosphate competitor and inhibits tropomyosin receptor tyrosine kinases (TRK) TRKA, TRKB, and TRKC, and also as proto-oncogene tyrosine-protein kinase ROS1 and anaplastic lymphoma kinase (ALK). TRK receptors produce cell proliferation through downstream signaling through the mitogen activated protein kinase, phosphoinositide 3-kinase, and phospholipase C-γ. ALK produces similar signaling with the addition of downstream JAK/STAT activation. Inhibition of those pathways suppresses neoplastic cell proliferation and shifts the balance in favor of apoptosis, resulting in shrinking of tumor volume. Literature survey revealed that there is no analytical methods have been reported individually or in combination with other drugs. This study describes a validated liquid chromatography method for Entrectinib in bulk and tablet dosage forms.

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MATERIALS AND METHODS

Chemicals and reagents

Entrectinib was procured from Hetero drugs Ltd, Hyderabad, India. HPLC grade water, methanol and Acetonitrile was obtained from Merck India Ltd.

Instrumentation:

Chromatographic separation of Entrectinib was achieved on Waters Alliance-e2695, using Symmetry ODS C- 18 (4.6 x 250mm, 5m) column and the mobile phase containing Methanol: Phosphate Buffer in proportion 35:65% v/v. The flow rate was 1.0 ml/min. An injection volume was 20 µl and the column temperature was 30°C. The runtime was 10.0 min.

Preparation of Potassium dihydrogen Phosphate (KH_2PO_4) buffer (pH-3.6):

Dissolve 6.8043 of potassium dihydrogen phosphate in 1000 ml HPLC water and adjust the pH 3.6 with diluted orthophosphoric acid. Filter and sonicate the solution by vacuum filtration and ultra-sonication.

Preparation of mobile phase:

Accurately measured 350 ml (35%) of Methanol, 650 ml of Phosphate buffer (65%) were mixed and degassed in digital ultra sonicater for 15 minutes and then filtered through µ filter under vacuum filtration.

Preparation of standard solution:

Accurately weigh and transfer 10 mg of Entrectinib working standard into a 10ml of clean dry volumetric flasks add about 7ml of Methanol and sonicate to dissolve and removal of air completely and make volume up to the mark with the same Methanol. Further pipette 0.1ml of the above Entrectinib stock solutions into a 10ml volumetric flask and dilute up to the mark with Methanol.

From the above stock solutions suitable aliquots were measured and transferred into individual volumetric flasks for obtaining the standards of varying concentrations ranging from 6 ppm to 14 ppm of Entrectinib.

Preparation of Sample Solution:

Weight 10 mg equivalent weight of Entrectinib sample into a 10 mL clean dry volumetric flask and add about 7mL of Diluent and sonicate to dissolve it completely and make volume up to the mark with the same solvent. Further pipette 0.1ml of Entrectinib above stock solution into a 10ml volumetric flask and dilute up to the mark with diluent.

Method development:

Initially the mobile phase tried was Methanol and Methanol: Water with varying proportions. Finally, the mobile phase was optimized to Methanol: Phosphate Buffer in proportion 35:65% v/v. The method was performed with various C18 columns like, X- bridge column, Xterra, and C18 column. Symmetry ODS C18 (4.6 x 250mm, 5m) was found to be ideal as it gave good peak shape and resolution at 1ml/min flow.

System Suitability

The standard solution was injected for five times and measured the area for all five injections in HPLC. The %RSD for the area of five replicate injections was found to be within the specified limits.

Assay

Inject the three replicate injections of standard and sample solutions and calculate the assay by using formula:

% ASSAY = Sample area/ Standard area × Weight of standard/Dilution of standard × Dilution of sample/ Weight of sample × Purity/ 100 × Weight of tablet/Label claim ×100

Method Validation

Linearity:

Inject each level of standards i.e. 6, 8, 10, 12, 14 and 16 ppm of Entrectinib into the chromatographic system and measure the peak area. Plot a graph of peak area versus concentration (on X-axis concentration and on Y-axis Peak area) and calculate the correlation coefficient.

Precision- Repeatability

The standard solution was injected for five times and measured the area for all five injections in HPLC. The % RSD for the area of five replicate injections was found to be within the specified limits.

Intermediate precision:

To evaluate the intermediate precision (also known as Ruggedness) of the method, Precision was performed on different days by maintaining same conditions Analyst- 1& 2,

Accuracy

Inject the Three replicate injections of individual concentrations (50%, 100%, 150%) were made under the optimized conditions. Recorded the chromatograms and measured the peak responses. Calculate the Amount found and Amount added for Entrectinib and calculate the individual recovery and mean recovery values.

RESULTS AND DISCUSSION

The method developed was validated as per ICH guidelines. The system suitability parmeters are performed and other parameters like Accuracy, Presicion, Linearty, Robustness, LOD, and LOQ are performed by using optimized chromatographic conditions.

System Suitability:

A standard solution was prepared by using Axitinib stock solution as per test method and was injected for 5 times into HPLC column. From the system suitability studies it is observed that % RSD values were within the specified limit i.e not more than 2. System suitability data was presented in table: 1

Table- 1: Results of system suitability for Entrectinib

S.no	Peak Name	RT	Area (µV*sec)	Height (µV)	USP Plate Count	USP Tailing
1	Entrectinib	2.277	1652847	185647	6589	1.24
2	Entrectinib	2.277	1653658	186254	6587	1.26
3	Entrectinib	2.267	1654521	185475	6584	1.28
4	Entrectinib	2.265	1653564	186594	6582	1.29
5	Entrectinib	2.277	1658745	185684	6895	1.24
Mean			1654667			
Std. Dev.			2355.764			
% RSD			0.142371			

Assay:

Under the optimized chromatographic conditions, the standard solution and the test solutions were injected in triplicate and the % Assay for the drug was calculated. The peak areas for the standards and test were given in the table -2

Table-2: Peak results for assay standard

S.no	Name	RT	Area	RT	Area
		Standard solution		Test Solution	
1	Entrectinib	2.265	1658254	2.246	1645879
2	Entrectinib	2.267	1658475	2.246	1645875
3	Entrectinib	2.267	1658471	2.246	1658423

Specificity:

It is the ability to asses unequivocally the analyte in the presence of components that may be expected to be present. Excipients that are commonly used were spiked into a pre weighed quantity of drugs. Appropriate dilutions were injected into chromatographic system and the quantities of the drugs were determined. The method was found to be specific with no interferences with blank

Linearity:

A series of solutions were prepared using Entrectinib working standard solution at a concentration levels from 6- 14 μg/ml of target concentration and the peak area response of all solutions are measured. A graph was plotted against the Concentration (μg/ml) on X-axis versus area/response on Y-axis. The detector response was found to be linear with a correlation coefficient of 0.999. Linearity graph is shown in Figure- 1 linearity results of the method are tabulated in table-3

Table-3: Data for Linearity of Entrectinib

Concentration (μg/ml)	Average Peak Area
6	1078475
8	1461129
10	1808358
12	2211573
14	2593778

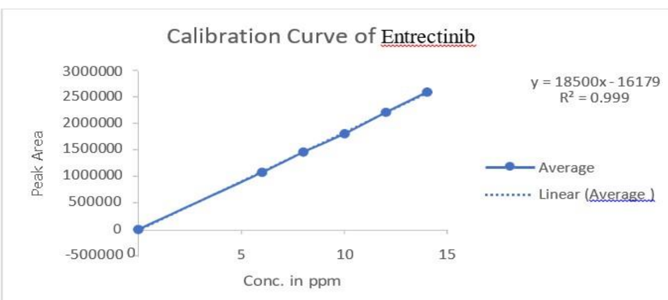

Fig-1: Linearity Curve of Entrectinib

Precision:

Precision studies were performed (Method, Day to day). The results are reported in term of Relative standard deviation. The repeatability studies were carried out by estimating response of 6 different concentrations of Entrectinib and reported in terms of % RSD, % RSD was 0.17, for repeatability and % RSD was 0.38 for Intermediate precision. The precision results of the method are tabulated in table-4

Table-4: Results of repeatability for Entrectinib

S.no	Peak Name	RT	Area (μV*sec)	Height (μV)	USP Plate Count
Repeatability			Intermediate Precision		
1	Entrectinib	2.293	1658954	2.274	1678541
2	Entrectinib	2.276	1658745	2.258	1685985
3	Entrectinib	2.286	1659865	2.267	1685745
4	Entrectinib	2.277	1653254	2.270	1685987
5	Entrectinib	2.280	1654781	2.264	1698526
6	Entrectinib	2.282	1654641	2.265	1685943
Mean			1657120		1686788
Std. Dev.			2913.592		6463.466
% RSD			0.175823		0.383182

Accuracy:

Accuracy of the method was determined by calculating the recovery of Entrectinib by the spiked method. Known quantity of Entrectinib was added to a pre-determined sample solution and the amount of Entrectinib was estimated by measuring peak areas. Mean % recovery values are within the limit (limit is 98-102%). Accuracy data was presented in table-5

Table-5: The accuracy results for Entrectinib

% Concentration	Area	Amt Added (ppm)	Amt Found (ppm)	% Recovery	Mean Recovery
50%	109068.3	5	5.021	100.420%	100.72%
100%	202187	10	10.054	100.540%	
150%	297032.3	15	15.181	101.206%	

SUMMARY

The analytical method was developed by studying different parameters. First of all, maximum absorbance was found to be at 235nm and the peak purity was excellent. Injection volume was selected to be 10μl which gave a good peak area. The column used for study was Symmetry ODS C18 (4.6×250mm, 5μm) because it was giving good peak. Ambient temperature was found to be suitable for the nature of drug solution. The flow rate was fixed at 1.0ml/min because of good peak area and satisfactory retention time. Mobile phase is Methanol: Phosphate Buffer pH-3.6 in the ratio of 35:65% v/v was fixed due to good symmetrical peak. So this mobile phase was used for the proposed study. Methanol was selected because of maximum extraction sonication time was fixed to be 10min at which all the drug particles were completely soluble and showed good recovery. Run time was selected to be 8min because analyze gave peak around 2.252minand also to reduce the total run time. The percent recovery was found to be 98.0-102 was linear and precise over the same range. Both system and method precision was found to be accurate and well within range. The analytical method was found linearity over the range of 6-14ppm of the Entrectinib target concentration. The analytical passed both robustness and ruggedness tests. On both cases, relative standard deviation was well satisfactory.

CONCLUSION

In the present investigation, a simple, sensitive, precise and accurate RP-HPLC method was developed for the quantitative estimation of Entrectinib in bulk drug and pharmaceutical dosage forms. This method was simple, since diluted samples are directly used without any preliminary chemical derivatisation or purification steps. Entrectinibwas found to be soluble in organic solvents such as ethanol, DMSO, and dimethyl formamide. Methanol: Phosphate Buffer (35:65) V/V was chosen as the mobile phase. The solvent system used in this method was economical. The %RSD values were within 2 and the method was found to be precise. The results expressed in Tables for RP-HPLC method was promising. The RP-HPLC method is more sensitive, accurate and precise compared to the Spectrophotometric methods. This method can be used for the routine determination of Entrectinibin bulk drug and in Pharmaceutical dosage forms.

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